# “NextGen Attendance Management System using Facial Recognition”

A Minor Project report submitted in partial fulfilment of the requirement for the award of the degree of

**BACHELOR OF TECHNOLOGY**

**IN**

**COMPUTER SCIENCE & ENGINEERING (CSE)**

**With**

**Artificial Intelligence & Machine Learning**

Submitted to

**SANDIP UNIVERSITY, SIJOUL**

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**MARCH, 2026**

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# DECLARATION

I hereby declare that the project report entitled **"NextGen Attendance Management System using Facial Recognition"** submitted in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering (CSE) With Artificial Intelligence and Machine Learning** my original work carried out under the guidance of **Dr. Syed Anas Ansar.**

I further declare that this project work has not been submitted elsewhere for the award of any other degree, diploma, associateship, fellowship, or any other similar title.

I also declare that wherever I have used materials such as data, theoretical analysis, or text from other sources, proper acknowledgment has been given and the sources have been duly cited.

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# APPROVAL CERTIFICATE

The project report entitled **"NextGen Attendance Management System using Facial Recognition"**, submitted by **"Kaushal Kumar Ray"** bearing **PRN No: 240205241023**, has been examined by us and is hereby approved for the award of the degree of **Bachelor of Technology in Computer Science and Engineering (CSE) With Artificial Intelligence and Machine Learning** at **Sandip University**.

It is understood that by this approval, the undersigned do not necessarily endorse or approve any statement made, opinion expressed, or conclusions drawn in this project report, but approve the project work only for the purpose for which it has been submitted.

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## ACKNOWLEDGEMENT


I would like to express my sincere and heartfelt gratitude to my project guide, **Dr. Syed Anas Ansar**, for his invaluable guidance, continuous support, and encouragement throughout the development of this project. His deep knowledge, constructive suggestions, and timely feedback helped me understand the concepts more clearly and significantly improved the overall quality of my work. I am truly grateful for his patience and constant motivation, which inspired me to complete this project successfully.

I would also like to express my sincere thanks to **Prof. Sonal Kumar**, Head of the Department of Computer Science and Engineering, for providing the necessary facilities and support to carry out this project work.

I am equally thankful to my friends and classmates for their cooperation, valuable discussions, and continuous support during the course of this project. Their ideas, feedback, and encouragement helped me overcome various technical and practical challenges.

I would also like to acknowledge the role of online resources, documentation, and research materials that contributed to the successful implementation of this project. These resources helped me explore new technologies and improve my understanding of deep learning and web development.

Finally, I would like to thank my friends and my parents for their continuous encouragement, motivation, and support during my academic journey at **Sandip University, Sijoul**.

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# ABSTRACT

Traditional attendance systems are inefficient, time-consuming, and prone to manipulation such as proxy attendance, which reduces reliability and increases administrative effort. Recent advancements in deep learning and computer vision have enabled the development of robust facial recognition systems for automated identity verification [3][4].

This project presents an intelligent attendance management system using deep learning-based facial recognition techniques. The proposed system utilizes models such as ArcFace and DeepFace to generate high-dimensional facial embeddings, which are compared using similarity metrics for accurate identification [1][3].

The system is designed to handle variations in lighting, pose, and facial expressions, which are common challenges in real-world environments [4][6]. A web-based application built using Flask enables real-time attendance tracking, while PostgreSQL ensures secure data storage.

Experimental results demonstrate improved accuracy and efficiency compared to traditional systems, aligning with recent research in AI-based attendance solutions [7][8]. The system is scalable, secure, and suitable for deployment in educational and organizational environments.

Keywords: Deep Learning, Facial Recognition, Face Embeddings, ArcFace, DeepFace, Computer Vision, Flask, PostgreSQL,

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# Chapter 1

# Introduction

Attendance management plays a crucial role in academic and organizational environments, yet traditional methods remain inefficient and prone to manipulation. With the rise of deep learning and computer vision, facial recognition offers a smarter and contactless alternative. This section sets the foundation by outlining the problem, defining clear objectives, and establishing the scope of the proposed system.

## *1.1 Background*

Attendance management is a fundamental administrative activity in educational institutions and organizations, directly influencing performance evaluation, discipline monitoring, and compliance tracking. Traditionally, attendance has been recorded manually through roll calls or maintained using paper-based registers. While simple, these methods are inefficient, time-consuming, and prone to human error and manipulation. Recent studies highlight that automated attendance systems using facial recognition significantly reduce manual effort and improve accuracy [5][7].

With the advancement of digital technologies, alternative systems such as RFID (Radio Frequency Identification), biometric fingerprint scanners, and smart cards have been introduced. However, these systems still present limitations such as hardware dependency, risk of proxy usage, hygiene concerns (in touch-based systems), and additional infrastructure costs.

Recent developments in deep learning, particularly in the field of computer vision, have enabled significant improvements in facial recognition systems. Deep learning-based face recognition systems have shown superior performance compared to traditional methods due to their ability to learn discriminative facial features automatically [3]. Techniques based on Deep Convolutional Neural Networks (DCNNs) can extract high-dimensional facial features and generate unique embeddings for accurate identity verification. These advancements make facial recognition a reliable, contactless, and scalable solution for automated attendance management. Embedding-based approaches such as FaceNet and ArcFace map facial images into high-dimensional feature spaces, enabling efficient similarity comparison [1][3].

This project proposes a web-based automated attendance system that leverages facial recognition technology to detect and identify individuals in real-time, thereby streamlining attendance processes while enhancing accuracy and security.

### *1.2 Problem Statement*

Traditional attendance systems are inefficient, labor-intensive, and highly susceptible to manipulation, particularly in the form of proxy attendance. Manual methods such as roll calls and paper-based registers consume a significant amount of time and effort, especially in large classrooms or organizations. These approaches often lead to human errors, misplaced records, and delays in maintaining and retrieving attendance data, ultimately reducing overall reliability and administrative efficiency.

Although some institutions have adopted digital solutions like RFID systems, biometric scanners, and smart cards, these methods only partially address the problem. They still rely on physical interaction or dedicated hardware, which increases operational costs and introduces new challenges such as device maintenance, hygiene concerns, and the possibility of misuse (e.g., card sharing or unauthorized access). Additionally, such systems may not be flexible enough to adapt to modern environments like remote or hybrid setups.

In this context, there is a clear need for a more advanced and intelligent attendance management system that leverages modern technologies. An ideal solution should be secure, efficient, and fully automated, minimizing human intervention while maximizing accuracy. It should eliminate proxy attendance through reliable identity verification, reduce manual workload for instructors and administrators, and ensure real-time data recording and access. Furthermore, the system should operate in a completely contactless manner, making it suitable for diverse environments, and should be scalable enough to handle a large number of users without performance degradation.

Therefore, there is a need for a secure, efficient, and automated attendance system that can:

- Eliminate proxy attendance
- Reduce manual effort
- Ensure accurate and real-time data recording
- Operate in a contactless and scalable manner

### *1.3 Objectives*

The primary goal of this project is to design and develop a smart attendance management system using facial recognition technology. The proposed system aims to improve efficiency, accuracy, and security compared to traditional attendance methods, which often rely on manual entry or physical verification.

This project focuses on developing an automated attendance system powered by deep learning-based facial recognition techniques. By leveraging advanced models such as DeepFace, the system is capable of identifying individuals based on unique facial features, thereby minimizing human intervention. One of the key objectives is to reduce the manual effort involved in recording and managing attendance while eliminating the possibility of proxy attendance through reliable identity verification.

In addition, the system aims to enhance recognition accuracy by utilizing robust facial embedding techniques and optimized matching algorithms. Another important objective is to ensure secure storage and management of attendance data within a centralized database system, enabling efficient retrieval and analysis. Furthermore, the project emphasizes the development of a user-friendly web interface using frameworks like Flask, ensuring seamless interaction between users and administrators.

### *1.4 Scope of the Project*

The scope of this project involves the development of a web-based attendance management system that automates the process of attendance marking using facial recognition technology. The system is designed to operate in real-time, enabling quick and accurate identification of individuals through a camera or webcam interface.

The proposed system includes functionalities such as real-time face detection and recognition, followed by automatic attendance marking upon successful identification. All attendance data is securely stored in a centralized database, ensuring efficient data management and easy retrieval for reporting and analysis purposes. The web-based platform provides an interface for users to access attendance records, while administrators are given control over monitoring attendance and managing user data.

Although the system is designed to be efficient and reliable, certain limitations must be acknowledged. The performance of facial recognition algorithms can be influenced by environmental conditions such as lighting variations, camera quality, and facial obstructions like masks or accessories. Additionally, the system requires initial dataset collection and model configuration to achieve optimal accuracy and performance.

### *1.5 Significance*

The NextGen Attendance System provides an efficient and automated solution to the limitations of traditional attendance methods. Manual systems are time-consuming and prone to errors such as proxy attendance, which reduces reliability. This project addresses these issues by using facial recognition technology to ensure accurate and contactless attendance marking.

By integrating deep learning models like ArcFace and FaceNet, the system improves recognition accuracy and minimizes chances of manipulation. It reduces manual effort for instructors and simplifies attendance management through automation.

The system is also scalable and user-friendly, as it is implemented as a web-based application with secure database storage. Features such as automatic attendance recording and leave management further enhance its usefulness.

Overall, the project demonstrates how AI-based solutions can improve efficiency, accuracy, and security in real-world applications like attendance management.

# Chapter 2
# Existing System and Literature Review

Current attendance systems still rely heavily on manual or semi-automated approaches, leading to inefficiencies and security concerns. At the same time, research in facial recognition has rapidly evolved through deep learning models. This part bridges real-world limitations with academic advancements, creating a strong base for the proposed solution.

## *2.1 Existing System*

Attendance management in most educational institutions is still largely dependent on traditional or semi-digital systems. The most common approach involves manual roll calls or maintaining physical registers, where attendance is recorded by instructors during class hours. While this method is simple, it is inefficient and consumes valuable instructional time.

In recent years, some institutions have adopted digital solutions such as RFID-based systems, biometric fingerprint scanners, and smart card technologies. These systems aim to automate attendance recording; however, they still rely heavily on physical interaction and additional hardware infrastructure. Moreover, they are not entirely secure, as RFID cards can be shared and fingerprint systems may face issues related to hygiene and device dependency.

The major problems observed in existing systems include:

- Long time required for attendance marking
- Dependency on manual or semi-manual processes
- Possibility of proxy attendance (e.g., card sharing)
- Lack of real-time tracking and analytics
- Increased administrative workload

## *2.2 Limitations of Existing System*

Despite technological advancements, current attendance systems still face several limitations that affect their efficiency and reliability. Manual and device-based systems often result in slow processing, as attendance must be recorded individually for each student, making them inefficient in large classrooms. These systems are also prone to proxy and manipulation, where students can bypass the system through methods such as RFID card sharing or manual marking.

In addition, many existing solutions depend on dedicated hardware like biometric scanners, which increases both cost and maintenance efforts. They also lack flexibility, especially in adapting to modern requirements such as remote or hybrid environments. Data management is another concern, as some systems still rely on fragmented or paper-based records, leading to inefficiencies in tracking and analysis. Furthermore, touch-based biometric systems raise hygiene concerns in health-sensitive situations.

These limitations highlight the need for a more advanced, automated, and contactless attendance solution.

### *2.3 Proposed Solution Overview*

To overcome the limitations of existing systems, this project proposes a deep learning-based automated attendance system using facial recognition integrated into a web application.

The proposed system leverages computer vision and deep learning techniques to detect and recognize faces in real-time using a camera feed. Once a face is identified, the system automatically marks attendance and stores the data securely in a centralized database. This eliminates the need for manual intervention and significantly reduces the possibility of proxy attendance.

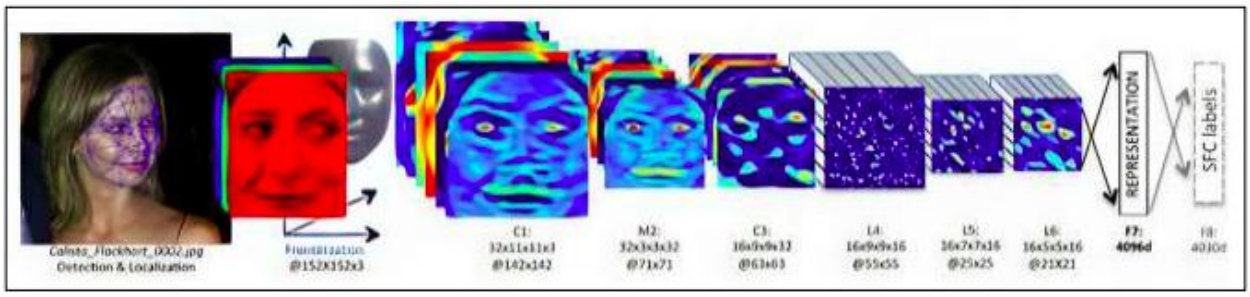



**Figure 1** Outline of the DeepFace Architecture [2]

The proposed system offers several important features, including contactless and automated attendance marking, real-time face detection and recognition, and high accuracy achieved through advanced deep learning models. It also ensures secure storage of attendance records and provides a web-based interface for easy access and management. Additionally, the system is scalable and can be effectively deployed in large classrooms or institutional environments.

Overall, this approach improves efficiency, enhances accuracy, and provides a better user experience compared to traditional attendance systems.

### *2.4 Literature Review*

Facial recognition has evolved from traditional image processing techniques to advanced deep learning-based approaches. Early systems relied on handcrafted features such as Haar cascades and LBPH, which suffered from poor performance under varying conditions [4].

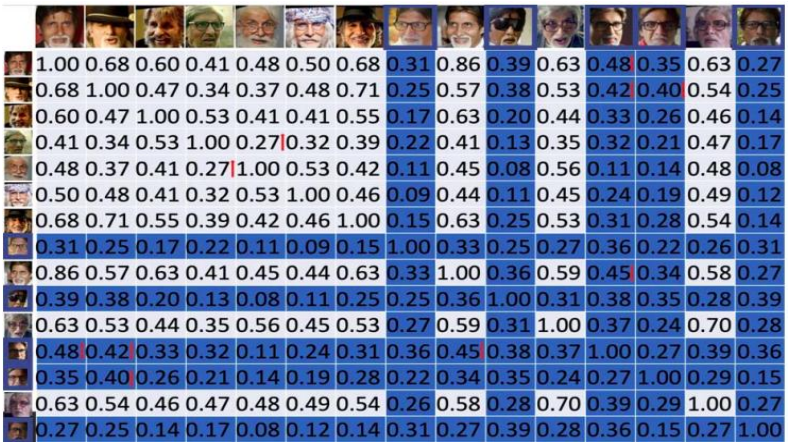
The introduction of deep convolutional neural networks (DCNNs) revolutionized face recognition by enabling automatic feature extraction and representation learning. Models such as FaceNet and DeepFace introduced embedding-based approaches, where facial images are mapped into a high-dimensional space for similarity comparison [2][3].

ArcFace further improved this concept by introducing an additive angular margin loss, which enhances intra-class compactness and inter-class separability, resulting in improved recognition accuracy [3]. This method has been widely adopted in real-world applications due to its robustness and efficiency [1].

Recent research focuses on applying these techniques to attendance systems. Studies such as Face-App and AI-powered attendance systems demonstrate the effectiveness of real-time facial recognition in educational environments [5][7]. These systems provide contactless, automated solutions that eliminate proxy attendance and improve administrative efficiency.

Additionally, modern systems incorporate advanced features such as cloud storage, real-time analytics, and AI-based decision-making modules [8][9]. However, challenges such as lighting variations, occlusions, and scalability still remain areas of ongoing research [6].

Overall, the literature highlights a clear shift toward deep learning-based facial recognition systems, forming the foundation for the proposed solution.



| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1.00 | 0.68 | 0.60 | 0.41 | 0.48 | 0.50 | 0.68 | 0.31 | 0.86 | 0.39 | 0.63 | 0.48 | 0.35 | 0.63 | 0.27 |
| 0.68 | 1.00 | 0.47 | 0.34 | 0.37 | 0.48 | 0.71 | 0.25 | 0.57 | 0.38 | 0.53 | 0.42 | 0.40 | 0.54 | 0.25 |
| 0.60 | 0.47 | 1.00 | 0.53 | 0.41 | 0.41 | 0.55 | 0.17 | 0.63 | 0.20 | 0.44 | 0.33 | 0.26 | 0.46 | 0.14 |
| 0.41 | 0.34 | 0.53 | 1.00 | 0.27 | 0.32 | 0.39 | 0.22 | 0.41 | 0.13 | 0.35 | 0.32 | 0.21 | 0.47 | 0.17 |
| 0.48 | 0.37 | 0.41 | 0.27 | 1.00 | 0.53 | 0.42 | 0.11 | 0.45 | 0.08 | 0.56 | 0.11 | 0.14 | 0.48 | 0.08 |
| 0.50 | 0.48 | 0.41 | 0.32 | 0.53 | 1.00 | 0.46 | 0.09 | 0.44 | 0.11 | 0.45 | 0.24 | 0.19 | 0.49 | 0.12 |
| 0.68 | 0.71 | 0.55 | 0.39 | 0.42 | 0.46 | 1.00 | 0.15 | 0.63 | 0.25 | 0.53 | 0.31 | 0.28 | 0.54 | 0.14 |
| 0.31 | 0.25 | 0.17 | 0.22 | 0.11 | 0.09 | 0.15 | 1.00 | 0.33 | 0.25 | 0.27 | 0.36 | 0.22 | 0.26 | 0.31 |
| 0.86 | 0.57 | 0.63 | 0.41 | 0.45 | 0.44 | 0.63 | 0.33 | 1.00 | 0.36 | 0.59 | 0.45 | 0.34 | 0.58 | 0.27 |
| 0.39 | 0.38 | 0.20 | 0.13 | 0.08 | 0.11 | 0.25 | 0.25 | 0.36 | 1.00 | 0.31 | 0.38 | 0.35 | 0.28 | 0.39 |
| 0.63 | 0.53 | 0.44 | 0.35 | 0.56 | 0.45 | 0.53 | 0.27 | 0.59 | 0.31 | 1.00 | 0.37 | 0.24 | 0.70 | 0.28 |
| 0.48 | 0.42 | 0.33 | 0.32 | 0.11 | 0.24 | 0.31 | 0.36 | 0.45 | 0.38 | 0.37 | 1.00 | 0.27 | 0.39 | 0.36 |
| 0.35 | 0.40 | 0.26 | 0.21 | 0.14 | 0.19 | 0.28 | 0.22 | 0.34 | 0.35 | 0.24 | 0.27 | 1.00 | 0.29 | 0.15 |
| 0.63 | 0.54 | 0.46 | 0.47 | 0.48 | 0.49 | 0.54 | 0.26 | 0.58 | 0.28 | 0.70 | 0.39 | 0.29 | 1.00 | 0.27 |
| 0.27 | 0.25 | 0.14 | 0.17 | 0.08 | 0.12 | 0.14 | 0.31 | 0.27 | 0.39 | 0.28 | 0.36 | 0.15 | 0.27 | 1.00 |

**Figure 2** Cosine Similarity Matrix by ArcFace[1]

The paper further highlights that facial recognition becomes particularly difficult when there are high intra-class variation (e.g., due to aging, makeup, or occlusion) and low inter-class variation (e.g., impersonation). ArcFace addresses this by generating more compact and well-separated feature embeddings, leading to improved accuracy. Additionally, techniques such as RetinaFace for detection and embedding optimization strategies further enhance system performance.

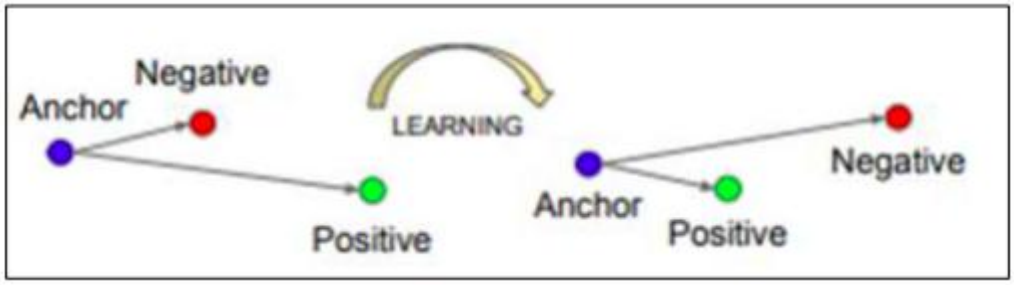



**Figure 3** The Triplet Loss [1][2]

Overall, the literature indicates a clear shift from traditional algorithms to deep learning-based approaches, with models like FaceNet, DeepFace, and ArcFace leading the field. These advancements provide the foundation for developing a robust, accurate, and scalable automated attendance system.

# Chapter 3

# System Overview

The overall structure of the system is shaped around automation, accuracy, and real-time performance. Key requirements, operational flow, and supporting technologies come together to define how the system functions in a practical environment.

## *3.1 System Analysis*

The proposed system is a web-based automated attendance platform that utilizes deep learning-based facial recognition combined with real-time processing. It replaces traditional attendance methods with a contactless, intelligent, and scalable solution. The system operates through multiple modules, including user registration, facial data processing, real-time recognition, attendance management, and leave handling. These components work together to ensure accurate identity verification while maintaining efficiency and usability.

### ***3.1.1 Functional Requirements***

Functional requirements define the essential operations that the system must perform to meet its objectives. The system is designed to support the complete attendance workflow, from user registration to attendance tracking and leave management.

The key functional capabilities of the system include:

- Student registration using name and unique ID
- Capturing and storing multiple facial images during registration
- Generating and storing facial embeddings for recognition
- Real-time face detection and identification using a camera feed
- Prevention of duplicate attendance entries for the same day
- Storage of attendance records with date and timestamp
- Retrieval of attendance history and reports
- Leave management based on attendance percentage with automated approval logic

### ***3.1.2 Non-Functional Requirements***

Non-functional requirements define the quality attributes and performance standards of the system. These ensure that the system operates efficiently, securely, and reliably under various conditions.

The system is designed to achieve:

- **Security**, by ensuring safe handling of user data and facial embeddings
- **Performance**, through real-time processing with minimal delay
- **Accuracy**, using deep learning models for reliable recognition
- **Scalability**, enabling support for multiple users simultaneously
- **Usability**, with a simple and intuitive user interface
- **Reliability**, ensuring consistent operation under different conditions
- **Maintainability**, allowing easy updates and future improvements

### ***3.2 Hardware Requirements***

The system requires moderate hardware resources to support real-time image processing and deep learning operations. Adequate hardware ensures smooth execution and reliable performance.

The minimum hardware specifications include:

- **Processor:** Intel i5 or higher
- **RAM:** Minimum 16 GB
- **Storage:** 512 GB (HDD/SSD)
- **Camera:** Webcam or external camera for face capture

These specifications enable efficient handling of facial recognition tasks and database operations.

### *3.4 Tools and Technologies*

The development of the system involves the integration of multiple technologies from the fields of computer vision, deep learning, and web development. Python with the Flask framework is used for backend development and API handling, enabling smooth communication between system components. OpenCV is utilized for real-time face detection and image preprocessing, while DeepFace, particularly using the ArcFace model, is employed for generating facial embeddings and performing recognition.

PostgreSQL, hosted on Neon, is used as the database for storing structured user and attendance data, ensuring reliability and scalability. Cloudinary is integrated for managing and storing captured facial images in the cloud, reducing local storage dependency. The frontend is developed using HTML, CSS, and JavaScript, providing an interactive and user-friendly interface. Together, these technologies enable the system to function as a robust, scalable, and efficient attendance management solution.

# Chapter 4
# System Design

Designing the system involves breaking it down into structured components that work seamlessly together. Architectural flow, data movement, user interactions, and database relationships are organized to ensure scalability and efficient processing.

## *4.1 System Architecture*

The proposed system follows a *modular web-based architecture* that integrates frontend, backend, and AI-based facial recognition components. The system is designed to ensure scalability, maintainability, and real-time performance.

The frontend interface allows users to interact with the system, while the Flask backend processes requests, performs facial recognition, and communicates with the database.

The architecture is divided into the following components:

- **Frontend Layer:** Built using HTML, CSS, and JavaScript for user interaction
- **Backend Layer:** Flask-based REST API handling logic and processing
- **AI Module:** DeepFace (ArcFace) for facial recognition
- **Database Layer:** PostgreSQL (Neon) for storing user and attendance data
- **Cloud Storage:** Cloudinary for storing facial images

Additionally, the backend is structured using *Flask Blueprints,* which modularize the application into components such as authentication, attendance, and leave management.

## *4.2 Data Flow Diagram (DFD)*

### *4.2.1 Level-0 DFD*

The Level-0 Data Flow Diagram (DFD) provides a high-level overview of the system by illustrating the flow of data between the main components: the user, the application, and the database. The user interacts with the system through a web interface to perform tasks such as registration, attendance marking, and submitting leave requests.

The application acts as the central processing unit, where inputs are validated and processed using appropriate logic. It then communicates with the database to store new information or

retrieve existing records. The database maintains all essential data, including user details, attendance logs, and leave information. This representation helps in understanding the overall system workflow in a simple and structured manner without going into internal details.

**User → Application → Database**

- The user interacts with the application through the web interface
- The application processes input (registration, attendance, leave requests)
- Data is stored and retrieved from the database

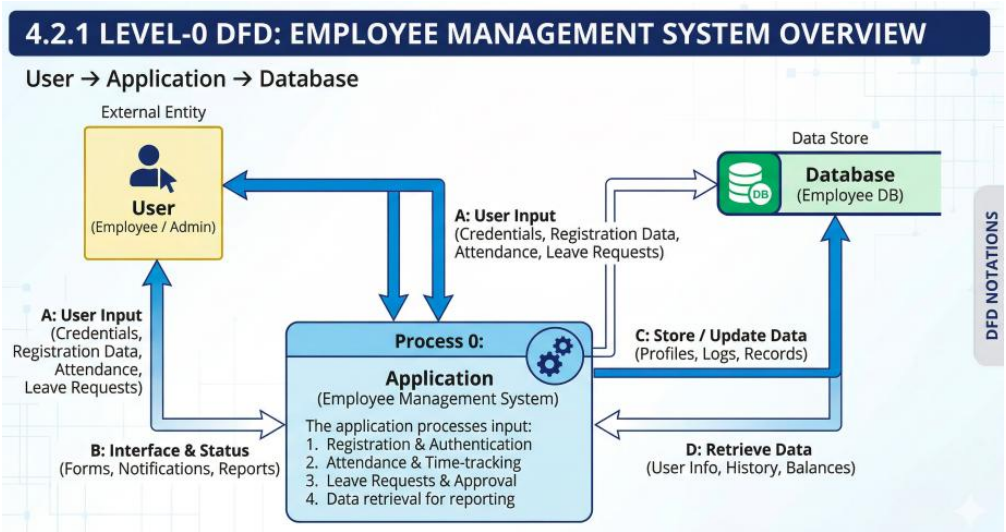



**Figure 4** Level- 0 DFD

### *4.2.2 Level-1 DFD*

The Level-1 Data Flow Diagram (DFD) provides a detailed view of the system by breaking it down into major functional modules. Each module is responsible for a specific set of operations, enabling better organization and efficient processing.

The User Module manages activities such as registration and login, ensuring proper user access. The Face Processing Module captures facial images and generates embeddings required for recognition. The Attendance Module records and maintains attendance data based on successful identification. The Leave Management Module handles leave requests and apply approval logic. The Database Module stores and manages all relevant data, including user details, attendance records, and embeddings.

- **User Module:** Handles registration and login

- **Face Processing Module:** Captures images and generates embeddings
- **Attendance Module:** Marks and stores attendance
- **Leave Management Module:** Handles leave requests and approvals
- **Database Module:** Stores user, attendance, and embedding data

Data flows between these modules through API calls and database queries, ensuring smooth system operation.

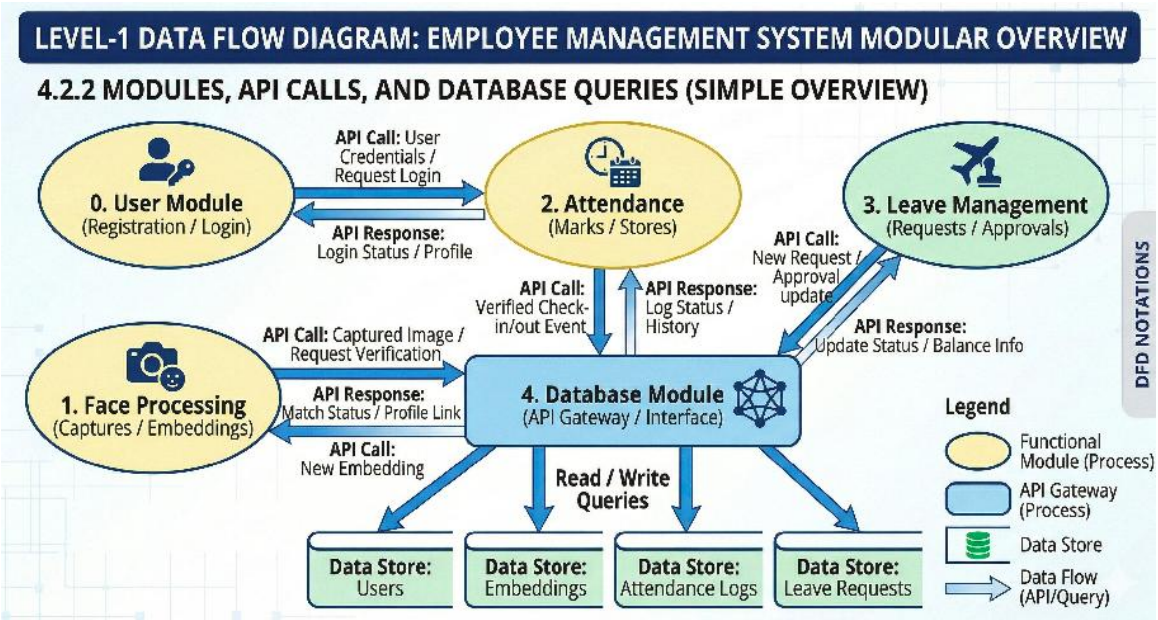



**Figure 5** Level-1 DFD

### *4.3 Use Case Diagram*

A Use Case Diagram is a behavioral diagram in UML that represents the interaction between users (actors) and the system. It provides a high-level view of system functionality by identifying different use cases and how external entities interact with them. The primary purpose of a use case diagram is to capture functional requirements and define the scope of the system in a simple and understandable way.

In this system, actors such as students and administrators interact with various use cases including registration, attendance marking, viewing records, and managing leave requests. Each use case represents a specific functionality offered by the system. The diagram helps in visualizing system behavior, improving requirement analysis, and ensuring that all user interactions are properly addressed during system design.

The use case diagram illustrates interactions between users and system functionalities, ensuring clarity in system behavior.

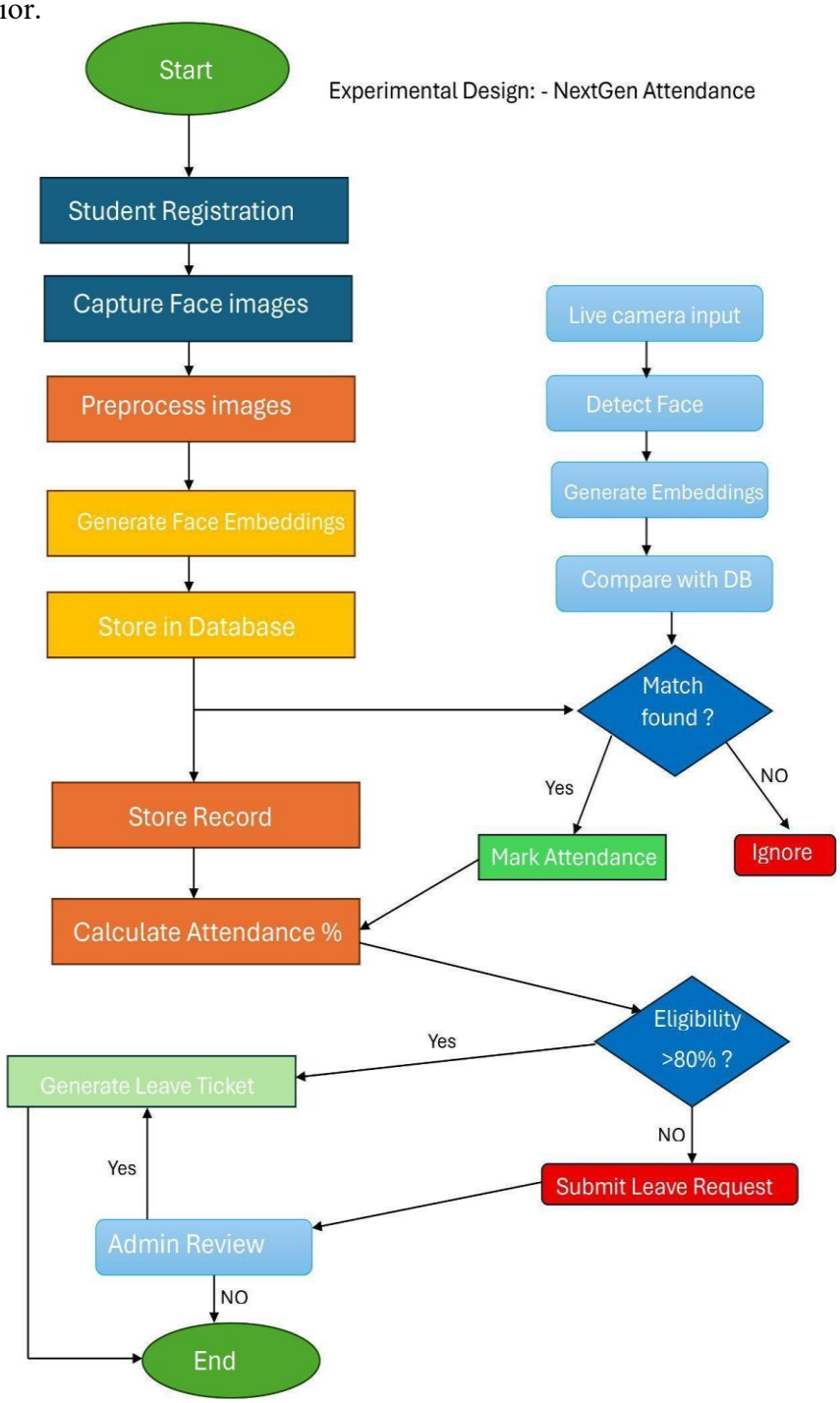



**Figure 6** System Architecture flow diagram

### *4.4 Database Design*

The system uses a relational database (PostgreSQL) to store structured data related to users, attendance, and leave management.

Entity-Relationship Overview, the main entities in the system include:

**Table 1** Table Information of the real-time database.

| Table Name | Description |
|---|---|
| Users | Stores student details such as ID, name, and embeddings |
| Attendance | Records attendance with date and timestamp |
| Leave | Stores leave requests and approval status |

#### ***4.4.1 Key Relationships***

- One **User** can have multiple **Attendance records**
- One **User** can submit multiple **Leave requests**
- Attendance is linked to users via unique ID

### *4.5 Methodology*

The use of embedding-based comparison aligns with modern deep learning approaches for face recognition, which have demonstrated high accuracy in real-world scenarios [3][4].

#### ***4.5.1 Registration Phase***

In this phase, students register by providing their name and unique ID. Multiple facial images are captured using a camera. These images are preprocessed using OpenCV techniques such as face detection and resizing.

Facial embeddings are then generated using DeepFace (ArcFace model), which converts facial features into numerical vectors. These embeddings, along with user details, are stored in the database.

This phase is implemented using:

- *capture_face()* → Captures dataset

- *generate_embeddings()* → Generates embeddings

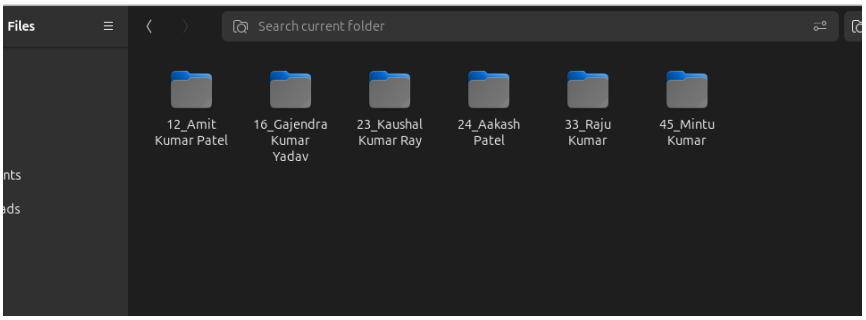



**Figure 7** A Snapshot of face database

**Figure 8** Detected and cropped faces

### ***4.5.2 Training Phase***

In the training phase, collected images are processed to generate embeddings for all registered users. These embeddings are stored in a serialized file (*embeddings.pkl*) for efficient access during recognition.

This step prepares the system for real-time face matching.

Implemented using: *train_model()*

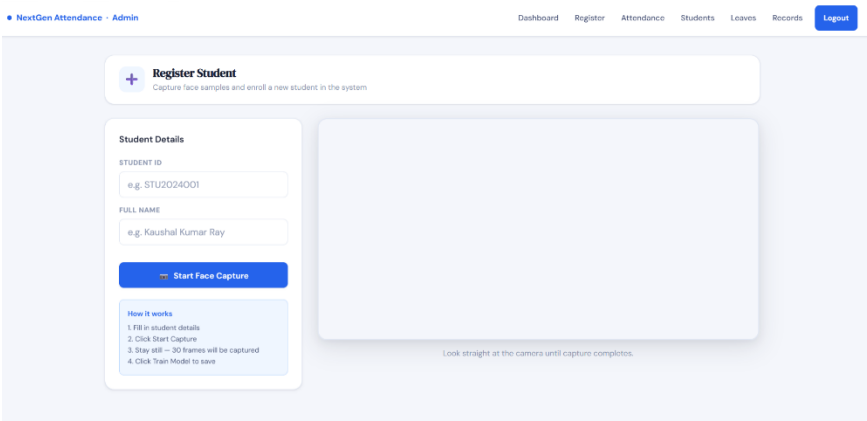



**Figure 9** Student Registration and Training model

### *4.5.3 Attendance Phase*

During this phase, the system captures live video input through a camera. Faces are detected using OpenCV, and embeddings are generated in real-time.

These embeddings are compared with stored embeddings using similarity metrics such as cosine distance. If a match is found, the system identifies the user and marks attendance automatically.

Implemented using: *process_attendance()*

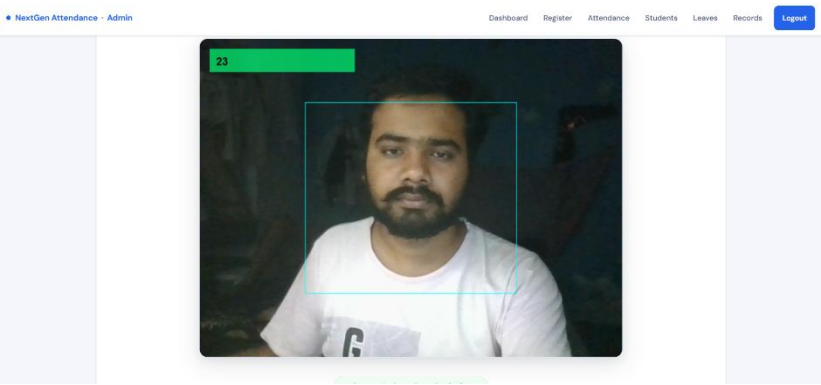



**Figure 10** Marking live attendance

### *4.5.4 Attendance Logic*

The system ensures accurate attendance management through the following logic:

- Attendance is stored with date and timestamp
- Duplicate entries are prevented for the same day
- Attendance history can be retrieved
- Attendance percentage is calculated dynamically

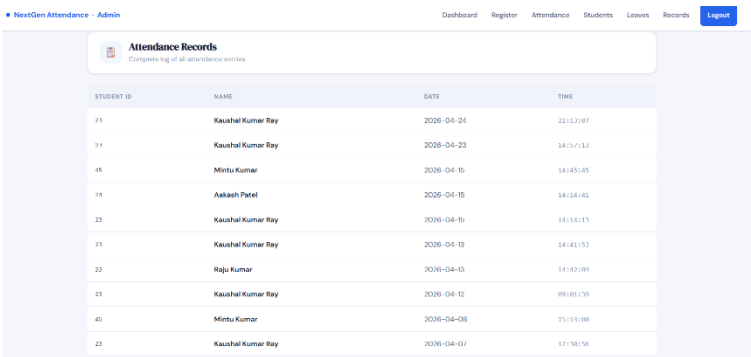



**Figure 11** Retrieval of the stored attendance from the database

### *4.5.5 Leave Management System*

The leave management module is integrated with attendance data. Leave approval is based on attendance percentage:

- If attendance ≥ 80% → Automatically approved
- If attendance < 80% → Marked as pending

Implemented in: *leave_routes.py*

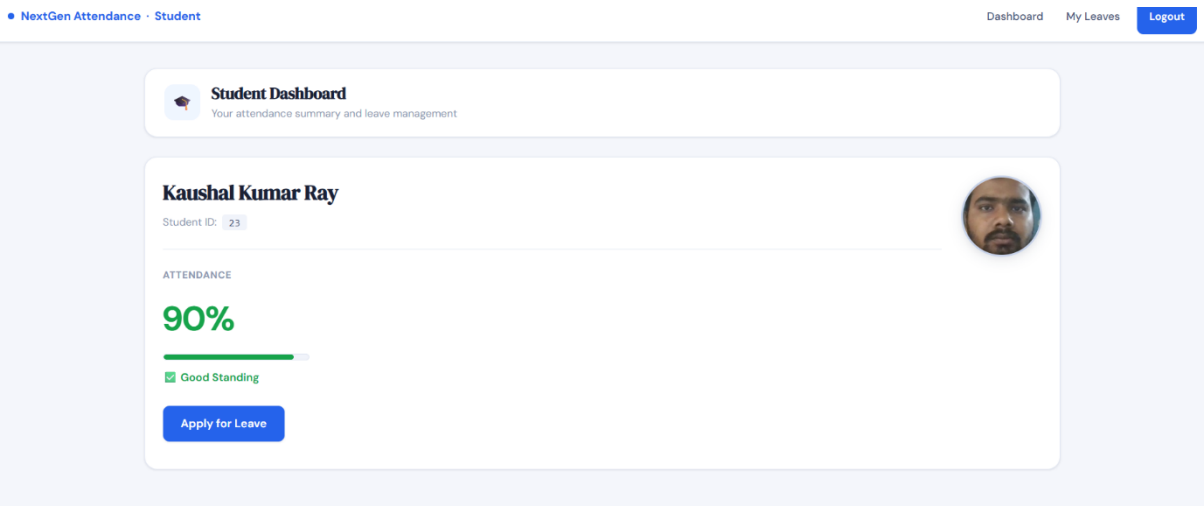



**Figure 12** Retrieval of the stored attendance from the database.

# Chapter 5

# System Implementation

Turning the design into a working solution involves integrating multiple technologies across frontend, backend, and database layers. Each module is developed to handle specific responsibilities, enabling real-time facial recognition and automated attendance management.

## *5.1 Frontend Implementation*

Turning the system design into a functional solution involves integrating multiple technologies across the frontend, backend, and database layers. Each module is designed to handle specific responsibilities, enabling real-time facial recognition and automated attendance management.

The frontend of the system is developed using HTML5, CSS, and JavaScript, providing a simple and user-friendly interface for interaction. The design emphasizes usability, clarity, and real-time feedback to enhance the overall user experience. It is responsible for managing user interactions such as student registration, attendance marking, and viewing attendance records. Communication between the frontend and backend is handled through HTTP requests using REST APIs.

The interface includes a clean layout for registration, attendance, and dashboard pages, along with form inputs for capturing student details such as name and ID. It also integrates camera functionality for capturing facial images through a webcam. The system provides real-time feedback by displaying messages indicating successful attendance marking or errors. CSS is used to improve the visual appearance, readability, and alignment of interface elements, while JavaScript handles dynamic functionalities such as capturing images, sending requests to the backend, and updating the user interface.

## *5.2 Backend Implementation*

The backend of the system is implemented using Python with the Flask framework, which manages the core application logic, API routing, and integration with artificial intelligence models and the database. Flask is organized using Blueprints, which allow modular separation of different components such as authentication, attendance management, and leave handling.

The backend is responsible for processing API requests received from the frontend, performing facial recognition using DeepFace, managing attendance logic and validation, and interacting with the PostgreSQL database. It also integrates with Cloudinary for storing captured facial images in a cloud-based environment.

Key backend components include the Flask application, which acts as the central server handling all requests and responses, and Blueprints, which organize the application into modular routes for better maintainability. REST APIs are used to enable seamless communication between the frontend and backend. DeepFace is integrated for generating facial embeddings and performing face matching, while OpenCV is used for real-time face detection and image preprocessing.

Example functional modules:

- *capture_face()* → Captures user facial images
- *generate_embeddings()* → Converts images into embeddings
- *process_attendance()* → Matches faces and marks attendance
- *leave_routes.py* → Handles leave logic and approval system

### *5.3 Database Implementation*

The system uses PostgreSQL *(Neon cloud database)* to store structured data in a reliable and organized manner. It ensures efficient storage, quick retrieval, and proper management of user information, attendance records, and leave details. The use of a relational database allows data to be stored in well-defined tables with relationships, improving consistency and reducing redundancy. Additionally, it supports secure data handling and enables smooth execution of queries required for real-time operations within the system.

**Table 2** Students Records Storage

| Field Name | Data Type | Description |
|---|---|---|
| Id | VARCHAR | Unique student ID (Primary Key) |
| name | VARCHAR | Name of the Student |
| Created_at | TIMESTAMP | Timestamp when the record was created (default: current time) |

**Table 3** Architecture of the attendance marking section

| Field Name | Data Type | Description |
|---|---|---|
| Id | SERIAL | Unique attendance record ID (Primary Key) |
| Student_id | VARCHAR | References student id (foreign Key) |
| date | DATE | Date of attendance |
| time | TIME | Time of attendance marking |

**Table 4** Leave request management

| Field Name | Data Type | Description |
|---|---|---|
| Id | SERIAL | Unique leave request ID (Primary Key) |
| Student_id | VARCHAR | References student ID |
| Reason | TEXT | Reason for leave |
| Status | VARCHAR | Leave status (default: 'Pending') |
| Created_at | TIMESTAMP | Timestamp of leave request |

### ***5.3.1 Database Operations***

The backend performs several essential database operations to ensure efficient data management and system functionality. These operations include insertion, retrieval, updating, and validation of records. New user details and attendance records are inserted into the database during registration and attendance marking. The system retrieves attendance history and user information whenever required for display or analysis. It also updates records, particularly in the leave management module, where the status of leave requests is modified based on predefined conditions.

In addition, the system incorporates validation mechanisms to maintain data integrity. Attendance is marked only if no entry exists for the current date, ensuring that duplicate records are prevented. Furthermore, attendance percentage is calculated dynamically using stored attendance data, enabling accurate tracking and decision-making within the system.

## ***5.4 Key Features Implementation***

The system includes several important features implemented using a combination of deep learning and web technologies.

### *5.4.1 Face Recognition-Based Attendance*

The system automates attendance marking using facial recognition technology. Faces are first detected from the live camera feed using OpenCV, after which DeepFace generates facial embeddings using models such as ArcFace or FaceNet. These embeddings capture unique facial features and are compared with stored data using similarity metrics like cosine distance. If a match is found within a defined threshold, the system identifies the user and automatically records attendance along with the current date and time.

```
127.0.0.1 - - [06/Apr/2026 18:13:21] "POST /process_attendance HTTP/1.1" 200 -
Distance: 2.6926543110909673
detected: 23
127.0.0.1 - - [06/Apr/2026 18:13:25] "GET /students HTTP/1.1" 200 -
127.0.0.1 - - [06/Apr/2026 18:13:26] "GET /static/css/style.css HTTP/1.1" 304 -
[DB SUCCESS] 23
```

**Figure 13** Attendance marking and updating in database (Neon).

### *5.4.2 Automated Attendance Management*

The system ensures accurate and efficient attendance tracking through automated processes. Attendance is recorded with precise timestamps to maintain proper logs, while duplicate entries for the same day are prevented to ensure data integrity. The system also allows retrieval of attendance history for monitoring and reporting purposes, and automatically calculates attendance percentage based on stored records.

### *5.4.3 Leave Management System*

A rule-based leave management system is implemented to automate the approval process and reduce manual intervention. The system calculates the attendance percentage of each user based on recorded data. If the attendance is greater than or equal to 80%, the leave request is automatically approved. If the attendance falls below 80%, the request is marked as pending for further review.

This approach minimizes administrative workload, ensures consistency in decision-making, and maintains fairness across all users. Additionally, the automated logic contributes to a more efficient and scalable system, enabling sustainable implementation with reduced operational overhead.

- Attendance percentage is calculated
- If ≥ 80% → Leave is auto-approved
- If < 80% → Leave remains pending

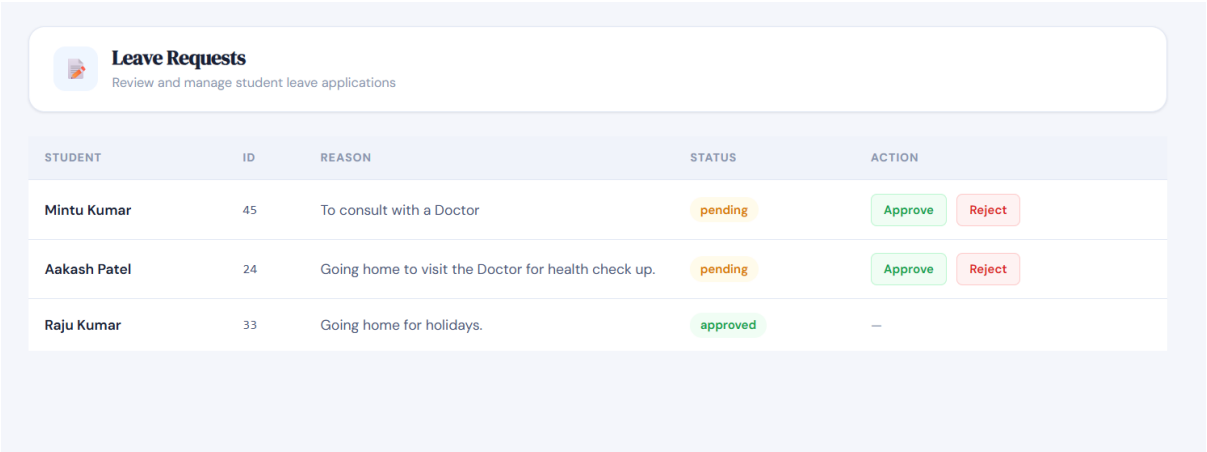



**Figure 14** Decision making dashboard leave requests.

This reduces administrative workload and ensures fairness. Also, this architecture enables sustainable implementation of the modern systems with less overheads.

### ***5.4.4 Cloud-Based Image Storage***

The system utilizes Cloudinary for storing captured facial images in a secure cloud environment. Images are uploaded directly to the cloud, reducing dependency on local storage and improving system scalability. This approach also enables easy access and efficient management of image data across the application.

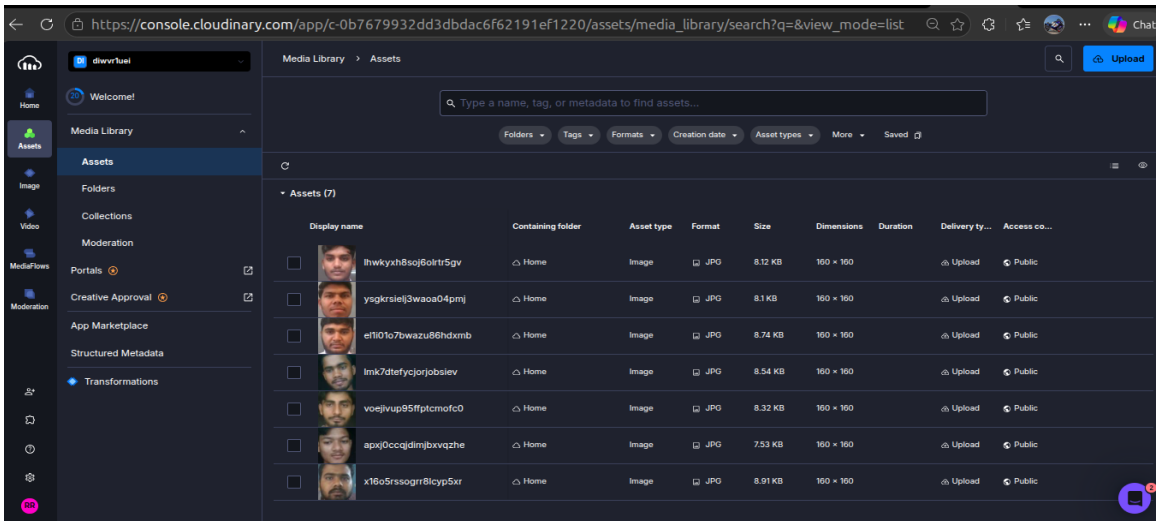



**Figure 15** Cloudinary assets dashboard where the facial data are stored.

### *5.4.5 Real-Time Processing*

The system supports real-time face detection and recognition by continuously processing live camera feeds. Faces are detected instantly using OpenCV, and facial embeddings are generated in real time using deep learning models. This ensures that the system can quickly identify individuals without noticeable delay.

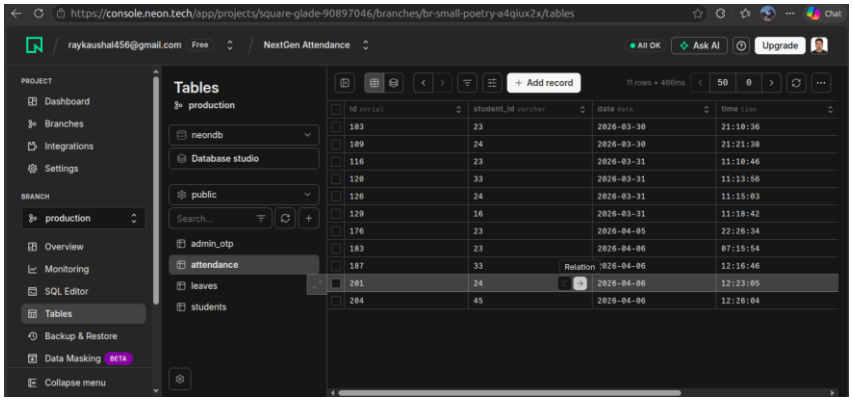



**Figure 16** Neon Database for real-time manipulation of the data.

The generated embeddings are then compared with stored data for rapid matching and identification. Once a match is found, attendance is marked immediately, allowing a seamless and efficient attendance process while maintaining accuracy and speed.

# Chapter 6
# System Testing

Ensuring reliability requires validating every part of the system under different conditions. Functional checks, performance evaluation, and real-world testing help confirm that the system behaves as expected and maintains accuracy.

## *6.1 Introduction to Testing*

System testing is a critical phase in software development that ensures the application functions correctly, efficiently, and reliably under different conditions. The proposed attendance system was tested to verify its functionality, accuracy, performance, and robustness.

## *6.2 Test Cases*

Test cases were designed to validate the correctness of individual functionalities. Each test case includes input conditions and expected outputs.

**Table 5** Functional Test Cases

| Test Case | Input | Expected Output |
|---|---|---|
| **User Registration** | Valid name and ID | User successfully registered |
| **Face Capture** | Camera enabled | Facial images captured and stored |
| **Embedding Generation** | Valid images | Embeddings generated successfully |
| **User Login (Admin)** | Correct credentials | Login successful |
| **User Login (Admin)** | Incorrect credentials | Error message displayed |
| **Face Recognition** | Registered face | User identified correctly |
| **Face Recognition** | Unregistered face | Access denied / no match |
| **Attendance Marking** | Recognized face | Attendance recorded with timestamp |
| **Duplicate Attendance Check** | Same user same day | Duplicate entry prevented |
| **Leave Request** | Attendance ≥ 80% | Leave auto-approved |
| **Leave Request** | Attendance < 80% | Leave marked as pending |
| **Database Storage** | Valid data | Data stored correctly in PostgreSQL |

### *6.3 Performance Testing*

Performance testing was conducted to evaluate the system's efficiency, responsiveness, and scalability under different conditions. The system's response time was measured for face detection, recognition, and attendance marking, and it demonstrated near real-time performance with minimal delay. Load handling was also tested by simulating multiple users and repeated requests, where the backend efficiently managed concurrent operations without significant performance degradation.

Additionally, processing speed was optimized by using precomputed embeddings (embeddings.pkl), reducing the time required for generating and comparing facial data. Database performance was evaluated to ensure that PostgreSQL could reliably store and retrieve attendance records with minimal latency, confirming the system's ability to handle regular operations effectively.

### *6.4 Testing Outcomes*

The testing phase confirmed that the system accurately identifies registered users and records attendance with proper timestamp validation. It effectively prevents duplicate entries and ensures that attendance is marked only once per day. The leave management logic also functions correctly based on attendance percentage.

Overall, the system performs efficiently under normal and moderate load conditions. Minor issues related to performance and accuracy were identified during testing and subsequently resolved, resulting in improved reliability and stability of the system.

# Chapter 7

# Results and Discussion

The implemented system demonstrates strong performance in recognizing faces and managing attendance automatically. Observations from outputs and system behavior provide insights into accuracy, efficiency, and practical usability.

## *7.1 Overview of Results*

The developed system was successfully implemented and tested for real-time attendance management using facial recognition. The results demonstrate that the system is capable of accurately detecting and recognizing faces, marking attendance automatically, and managing attendance records efficiently.

The integration of deep learning models significantly improved recognition accuracy compared to traditional methods. The performance of the proposed system is consistent with recent AI-based attendance systems, which report high accuracy and real-time efficiency [7][8].

## *7.2 System Outputs*

The system generates multiple outputs during its operation, reflecting the different functionalities of the application. During user registration, the system provides confirmation after successful registration and embedding generation. In the face recognition phase, it identifies the user in real time and displays the recognized name or ID.

The system automatically records attendance along with the corresponding date and timestamp, ensuring accurate tracking. It also manages leave requests by displaying approval or pending status based on the user's attendance percentage. Additionally, the dashboard presents a summary of attendance records and history, allowing both users and administrators to monitor performance effectively.

## *7.3 Screenshots and System Interface*

The system interface is designed to be simple, intuitive, and user-friendly, ensuring ease of use for both students and administrators. It provides clear navigation and organized layouts, enabling users to perform tasks such as registration, attendance marking, and record viewing efficiently.

### *7.3.1 Major Screens in the System:*

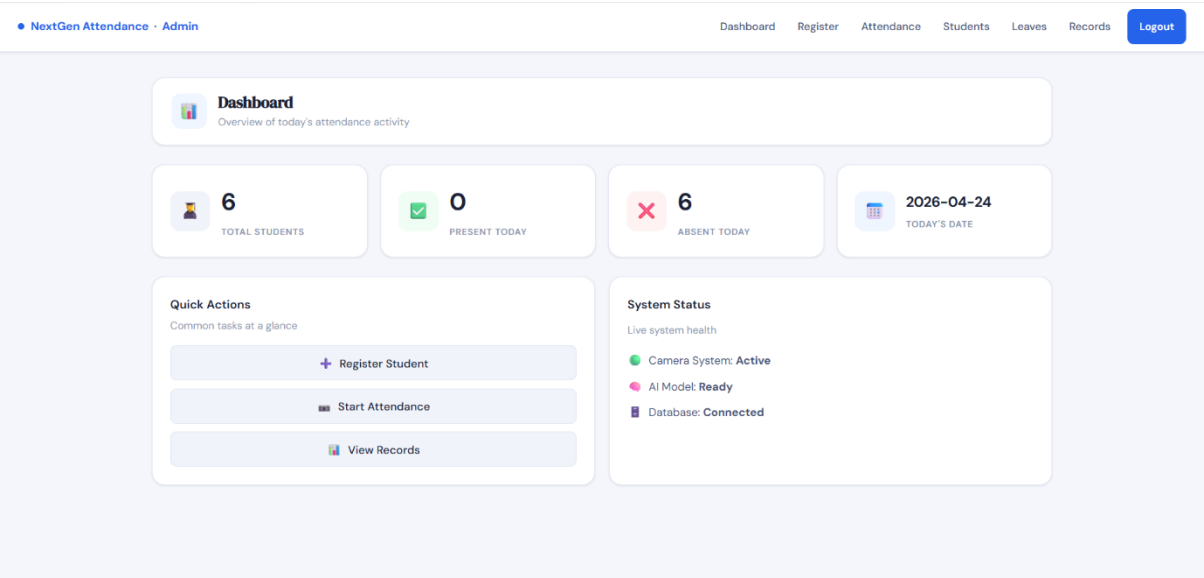



**Figure 17** Homepage Dashboard.

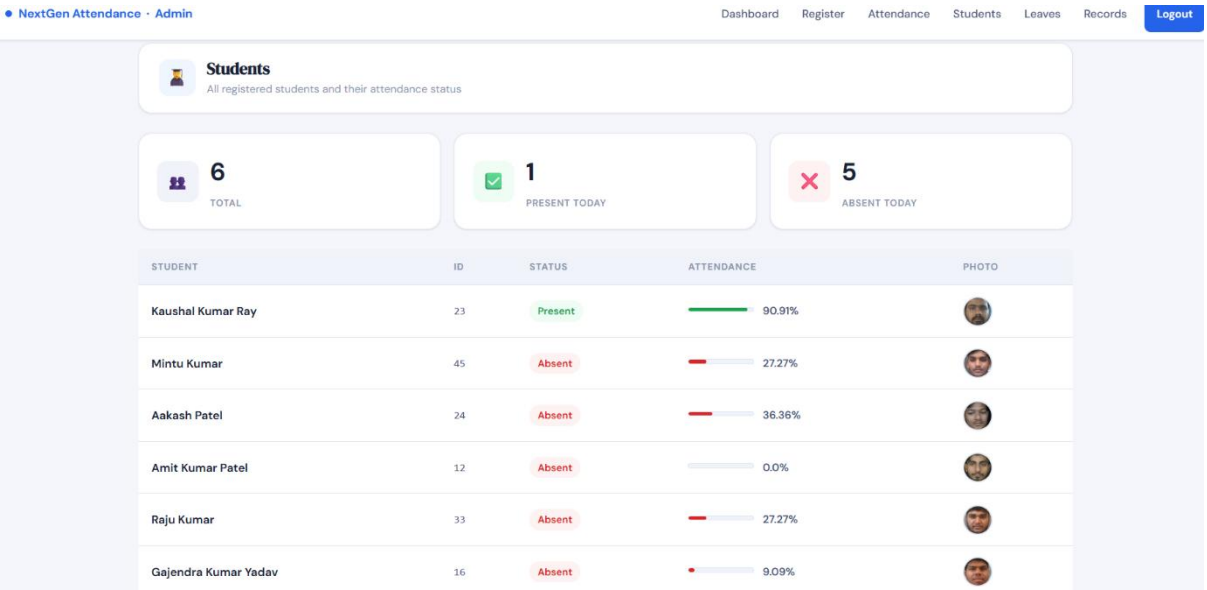



**Figure 18** Attendance dashboard displaying records.

### *7.4 Performance Analysis*

The system was evaluated based on key factors such as accuracy, speed, reliability, and scalability. Recognition accuracy was significantly improved through the use of DeepFace with the ArcFace model, while capturing multiple images per user further enhanced the overall performance. The system performs face recognition and attendance marking in near real-time, ensuring minimal delay during operation.

In terms of reliability, the system consistently identifies users under standard conditions and effectively prevents duplicate attendance entries. Additionally, the system demonstrates good scalability by handling multiple users efficiently, making it suitable for deployment in institutional environments.

### *7.5 Discussion*

The results indicate that the proposed system effectively addresses the limitations of traditional attendance systems. By leveraging deep learning-based facial recognition, the system eliminates manual intervention and reduces the possibility of proxy attendance.

The use of ArcFace embeddings significantly improves recognition reliability, especially under variations in facial appearance [1][3]. However, certain challenges such as lighting conditions and occlusions can still impact accuracy, which can be further improved with enhanced datasets and model tuning.

Overall, the system demonstrates strong potential for real-world deployment in educational institutions and organizations.

### *7.6 Summary*

The system successfully achieves its objectives by providing an automated and intelligent approach to attendance management using facial recognition technology. By eliminating manual attendance methods, the system ensures a seamless and contactless process where attendance is marked automatically through real-time face detection and recognition. This significantly reduces the time and effort required by instructors and administrators.

The use of deep learning models such as ArcFace improves the accuracy and efficiency of the system by enabling reliable identification of individuals even under varying conditions. As a result, issues like proxy attendance are effectively minimized, enhancing the integrity of attendance records. The system also reduces human errors associated with manual entry and ensures consistent data recording with proper timestamps.

Furthermore, the implementation of a web-based architecture makes the system scalable and user-friendly. It can handle multiple users efficiently and provides easy access to attendance records and reports. Overall, the system offers a modern, efficient, and reliable solution for attendance management in educational institutions and organizations.

# Chapter 8

# Conclusion and Future Work

The project successfully delivers an intelligent and automated attendance solution using facial recognition. While the current system achieves its goals, there remains scope for enhancements that can further improve accuracy, scalability, and real-world adaptability.

## *8.1 Conclusion*

The proposed NextGen Attendance System provides an efficient, intelligent, and automated solution for attendance management using deep learning-based facial recognition. The system successfully eliminates the need for manual attendance marking and significantly reduces the possibility of proxy attendance.

The implementation of facial recognition using advanced models such as ArcFace and FaceNet enables accurate identification of individuals in real-time. The system effectively captures facial data, generates embeddings, and compares them with stored records to mark attendance automatically.

During development, several challenges were encountered, including incorrect attendance logic, database connectivity issues, low recognition accuracy, UI limitations, and performance bottlenecks. These challenges were systematically addressed through improvements such as date-based attendance validation, optimized backend processing, enhanced UI design, and the use of more robust deep learning models.

The final system demonstrates:

- Accurate real-time face detection and recognition
- Automated attendance marking with timestamp validation
- Improved accuracy using deep learning and data augmentation
- Efficient database management and record handling
- A user-friendly web interface

Overall, the system proves to be scalable, reliable, and suitable for deployment in educational institutions and organizations, offering a modern alternative to traditional attendance systems.

### *8.2 Future Work*

Although the system performs effectively, there are several opportunities to further enhance its capabilities and usability. Future improvements can focus on expanding accessibility, strengthening security, and improving overall performance.

One important enhancement is the development of a mobile application, which would allow users and administrators to access the system more conveniently from anywhere. Security can be improved by implementing liveness detection techniques to prevent spoofing attempts using photos or videos. The accuracy of the system can also be enhanced by training models on larger and more diverse datasets, enabling better performance under challenging conditions such as poor lighting or occlusions.

In addition, deploying the system on cloud infrastructure would improve scalability and allow remote access across multiple locations. The inclusion of a real-time analytics dashboard can provide visual insights into attendance trends and performance. Support for multi-camera systems can enable simultaneous attendance tracking in multiple classrooms, making the system more suitable for large institutions. Furthermore, integrating the system with existing ERP or academic management systems would allow centralized data management and streamline administrative processes.

Overall, these enhancements would make the system more robust, scalable, and adaptable to real-world deployment scenarios.

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# Appendix

## A: Project Repository

The complete source code of the project is available on GitHub for reference and evaluation.

Repository Link:
https://github.com/Kaushal-Kumar-Ray/NextGen_Attendance

QR Code:

The repository contains:

- Backend implementation using Flask
- Face recognition modules (DeepFace, ArcFace)
- Database schema and API routes
- Frontend interface and integration

## Appendix B: API Endpoints

The system exposes RESTful APIs for communication between frontend and backend.

| Endpoint | Method | Description |
|---|---|---|
| /Register | POST | Register new user |
| /attendance | POST | Mark attendance |
| /apply-leave | POST | Apply leave |
| /attendance | GET | Fetch attendance records |
| /login | POST | Admin authentication |

## Appendix C: Setup Instructions

To run the system locally:

1. Clone the repository
   git clone https://github.com/Kaushal-Kumar-Ray/NextGen_Attendance
2. Install dependencies
   pip install -r requirements.txt
3. Configure environment variables (.env file)
4. Run the Flask server
   python app.py
5. Open browser
   http://localhost:5000

## Appendix D: Personal Experience and Reflections

Working on the NextGen Attendance Management System provided valuable hands-on experience in developing a real-time AI-based application. One of the main challenges was achieving accurate face recognition under varying conditions such as lighting and facial angles, which was improved through experimentation and threshold tuning. Integrating multiple technologies like Flask, DeepFace, PostgreSQL, and Cloudinary helped in understanding full-stack system design and deployment. The project also highlighted the importance of optimization, data quality, and secure handling of user information. Overall, this experience enhanced my practical knowledge of deep learning, computer vision, and system integration while improving my problem-solving skills.